

The Even Better Sinewave Indicator

“This is heavy stuff,” said Tom lighty.

I first introduced the Sinewave Indicator as early as 1996.^{1,2} The concept was based on the idea that no technical indicator is predictive. All indicators rely on historical data, and are therefore their signals are always lagging. However, cycles exist in the market, and it is often true that an observed cycle will continue for a short while into the future. Therefore, if we can transfer the cyclic data swings to a sine wave, we can then artificially advance the phase of the sine wave to create a predictive indicator.

Two conditions must be satisfied to make the Sinewave Indicator useful. First, the market must be in a cycle mode. Making a cyclic prediction when the data are in a trend is basically futile. Second, the period of the dominant cycle must be estimated with reasonable accuracy. The downside of the Sinewave Indicator is that the amplitude swings must be from -1 to $+1$, and therefore there is no direct indication of when the market is in a trend mode.

Since the roofing filter gives control of the frequency content of the data to be analyzed, the basic idea is to use the frequency content to an advantage. This control enables the Even Better Sinewave Indicator to not only enter a trade as quickly as possible, but also use the trend component to give the confidence to stay with an existing trade without being whipsawed.

■ Even Better Sinewave Approach

The original Sinewave Indicator was created by seeking the dominant cycle phase angle that had the best correlation between the price data and a theoretical dominant cycle sine wave. The Even Better Sinewave Indicator skips all the cycle measurements completely and relies on a strong normalization of the waveform at the output of a modified roofing filter. The modified roofing filter uses a single-pole high-pass filter to deliberately retain the longer-period trend components. The single-pole high-pass filter basically levels the amplitude of all the cycle components that would otherwise be larger with longer wavelengths due to Spectral Dilation. Therefore, when the waveform is normalized to the power in the waveform over a short period of time, the longer wavelength contributions tend to be an indication to stay in a trade when the market is in a trend.

The Even Better Sinewave Indicator works extraordinarily well when the market is in a trend mode. This means that the spectacular failures of most swing wave indicators are mitigated when the expected price turning point does not occur.

Although I have not studied it extensively, it appears that the Even Better Sinewave Indicator works well on futures intraday data. It takes a position in the correct direction and tends to stay with the good trades without excessive whipsawing.

■ Even Better Sinewave Description

The Even Better Sinewave Indicator is described with reference to the EasyLanguage code given in Code Listing 12-1. The only user input for the indicator is the expected maximum duration of the trades when the market is in a trend. Making the duration input parameter shorter will increase the number of trades over a given time span, and will have a minor effect of entering the trades a little sooner. Conversely, increasing the duration input parameter has the primary effect of holding and indicated trade position longer.

After declaring variables, the data is filtered in a single-pole high-pass filter. A single-pole high-pass filter is deliberately selected to equalize the data spectrum and allow trending components through the filter into the indicator. The degree of the spectrum allowed through the filter is established by the duration input. The value of the duration input is set to be approximately

the length of a trade position in a continuing trend. The default value of the duration input is 40 bars, so a maximum trade duration of about two months can be expected with this setting. Increasing the duration input will increase the maximum duration of a trade in a trend. This has the implication that you will be working through some additional drawdowns throughout the trade with the benefit that you will not be whipsawed out of a profitable trending trade.

The high-pass filtered data is then filtered in a SuperSmoother filter whose critical period is set to 10 bars.

After the data is passed through the modified roofing filter the three-bar average of the wave amplitude and the power are computed. Power is proportional to the square of the wave amplitude. The Even Better Sinewave is then computed by normalizing the averaged wave amplitude to the square root of the averaged power, causing the indicator to swing between -1 and $+1$.

Code Listing 12-1. Even Better Sinewave Indicator EasyLanguage Code

```
{
    Even Better Sinewave Indicator
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}

Inputs:
    Duration(40);

Vars:
    alpha1(0),
    HP(0),
    a1(0),
    b1(0),
    c1(0),
    c2(0),
    c3(0),
    Filt(0),
    count(0),
    Wave(0),
    Pwr(0);
```

(Continued)

```
//HighPass filter cyclic components whose periods are
shorter than Duration input
alpha1 = (1 - Sine (360 / Duration)) / Cosine(360 /
Duration);
HP = .5*(1 + alpha1)*(Close - Close[1]) + alpha1*HP[1];
//Smooth with a Super Smoother Filter from equation 3-3
a1 = expvalue(-1.414*3.14159 / 10);
b1 = 2*a1*Cosine(1.414*180 / 10);
c2 = b1;
c3 = -a1*a1;
c1 = 1 - c2 - c3;
Filt = c1*(HP + HP[1]) / 2 + c2*Filt[1] + c3*Filt[2];
//3 Bar average of Wave amplitude and power
Wave = (Filt + Filt[1] + Filt[2]) / 3;
Pwr = (Filt*Filt + Filt[1]*Filt[1] + Filt[2]*Filt[2]) / 3;
//Normalize the Average Wave to Square Root of the Average
Power
Wave = Wave / SquareRoot(Pwr);

Plot1(Wave);
```

■ Using the Even Better Sinewave Indicator

The Even Better Sinewave Indicator is shown in Figure 12.1, using the default duration setting of 40 bars. This means that the maximum length of a trade when the market is in a trend is about two months. This is because the high-pass filter starts to attenuate the cyclic components that are longer than 40 bars.

Interpretation of the Even Better Sinewave is simple. Hold a long position when the indicator value is near +1, and hold a short position (or go flat if trading to the long side only) when the indicator value is near -1.

The trade duration in a trend can be extended just by increasing the duration input value. For example, the indicator using a duration value of 40 is shown in Figure 12.2. Sure enough, the indicated long position in the uptrend is extended. Doing this causes some of the short-term reversals during the trend to be eliminated. The duration input parameter is adjusted to fit your trading style.

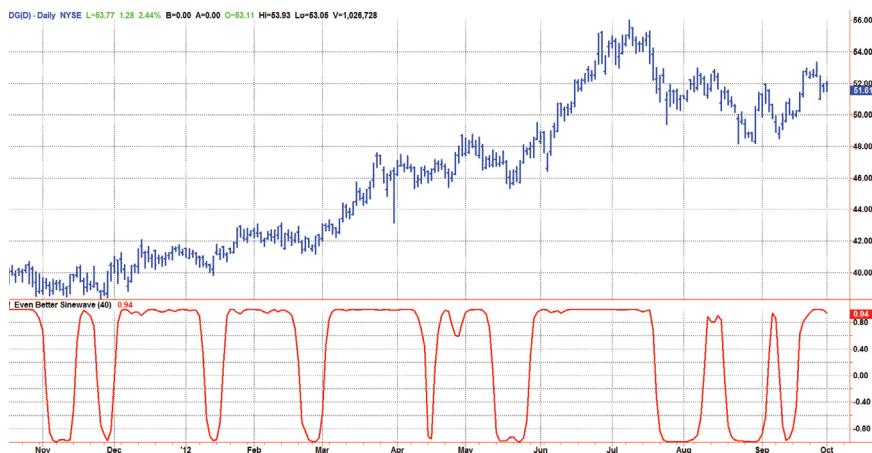


FIGURE 12.1 The Default Even Better Sinewave Indicator Holds a Position for about Two Months in a Trend

Conversely, decreasing the duration input parameter shortens the maximum trade length when the market is in a trend, being more sensitive to the shorter wavelengths in the data. For example, Figure 12.3 shows the indicator when the duration input is set to 20 bars. Some care should be taken when shortening the duration input because the resulting shorter trades are more sensitive to entry and exit timing and computational lag of the indicator.

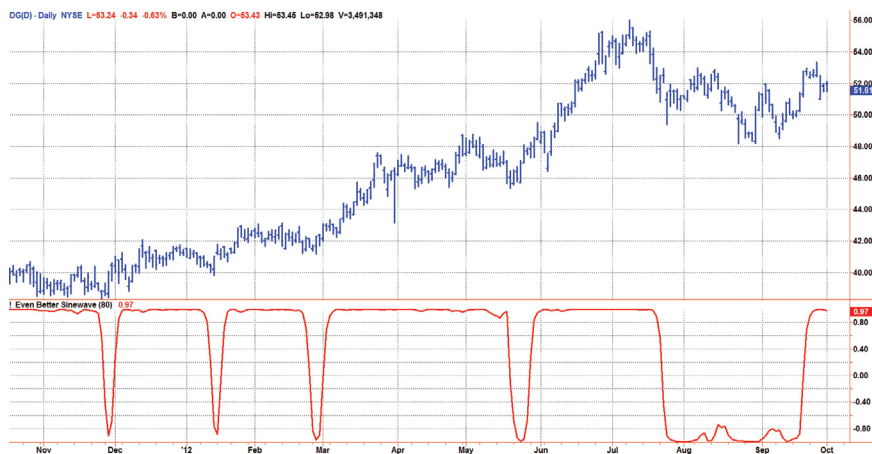


FIGURE 12.2 Trade Position Length is Increased by Increasing the Duration Input Parameter

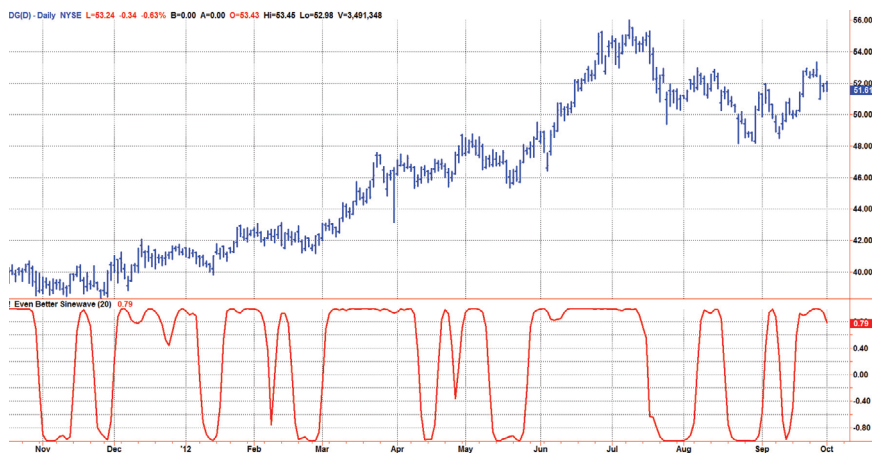


FIGURE 12.3 Decreasing the Duration Input Parameter Make the Even Better Sinewave Indicator More Sensitive to Short-Term Variations in the Data

■ Key Points to Remember

1. The Even Better Sinewave Indicator is a variant of the roofing filter, using a single-pole high-pass filter.
2. The unambiguous signals of the Even Better Sinewave Indicator are generated by normalizing the wave amplitude to the square root of the power.
3. The only input for the Even Better Sinewave Indicator is the duration parameter.
4. The duration parameter controls the maximum length of an indicated position by setting the critical period of the high-pass filter.

Notes

1. John F. Ehlers, “Stay in Phase,” *Stocks & Commodities*, November 1996, Vol. 14, No. 11, p. 69.
2. John F. Ehlers, *Rocket Science for Traders: Digital Signal Processing Applications* (Hoboken, NJ: John Wiley & Sons, 2001), Chapter 9.